

## Reference excerpt 02-009

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the two ECG and JVP traces (see Fig. 1). The non-invasive techniques used up to now, are based on the use of a microphone or a motion sensor and produce a trace which qualitatively represents the pressure in the IJV.[Pyhel(1978), Applefeld(1990), Mackenzie(1902), Mackay(1947), Kalmanson et al. (1972)] This new technique is different from those used previously in that the JVP is the instantaneous value of the CSA of the IJV. The US technique for the JVP represents an improvement from qualitative to quantitative JVP. The importance of having a quantitative JVP trace had already emerged in the past when a calibrated JVP has been used.[Pyhel(1978)] This new technique goes beyond the limit of calibration because it makes it possible to extrapolate the numeric parameters from the JVP that can be used for the clinical evaluation inter and intra-patient. This is a methodology-note that describes the innovative methods developed by the author to obtain quantitative parameters obtained by the JVP with ultrasound technique. The results reported here are only illustrative and are not produced by a dedicated experiment. 1 METHODOLOGY PRESENTATION ECG analysis and P, R, T events detection Over the past 30 years, several algorithms for the automatic detection of structures P, QRS and T in ECG, have been developed[Kohler(2002)]. However, these can also be analysed manually. In this work, the video clip of the IJV was analyzed frame by frame to find the sonograms in which the cursor of the ECG was at one of the events P, R or